Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

Section B **7*3=21**

Section C **7*4=28**

Section D **7*3=21**

19)	Given a set of cities connected by roads with varying distances between them, formulate this as a graph and use the concept of a minimum spanning tree to plan an efficient route for a delivery service.	**7**	(CO3, L4)
20)	Prove that if G is a graph with more than one vertex and the degree of each vertex in G is at least n/2 (where n is the number of vertices), then G contains a Hamiltonian path.	**7**	(CO2, L4)
21)	Given a graph with 5 vertices and 6 edges, prove that the graph is not planar by applying Kuratowski's theorem.	**7**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use